CAUSE & COUNTERMEASURE DIAGNOSTIC

PURPOSE

Separate evidence about why a condition occurred from evidence that an intervention changes the outcome. Make the reasoning visible before concluding that a successful action proves the cause.

WHAT IT IS

- A practitioner diagnostic
- A way to separate causal reasoning from intervention evidence
- A structure for making assumptions and uncertainty visible
- A starting point for stronger investigation and review

WHAT IT IS NOT

- A prescribed root cause methodology
- A scoring or maturity model
- Proof that no recurrence means the cause was correct
- A substitute for professional judgment

DIAGNOSTIC LENS

A. START WITH THE OBSERVED CONDITION

Describe what actually happened before explaining why. Keep observation separate from interpretation.

What condition, event, or outcome was observed?

What evidence establishes that it occurred?

What do we know about when, where, and under what conditions it occurred?

CAUSAL REASONING *Why did it happen?*	INTERVENTION *What did we change?*
01 HYPOTHESIS *What explanation are we proposing for why the condition occurred?*	**04 COUNTERMEASURE** *What did we change, remove, add, contain, or control?*
02 EVIDENCE *What evidence supports the proposed causal relationship?*	**05 MECHANISM** *How is the intervention expected to change the outcome?*
03 ALTERNATIVES *What other explanations remain plausible?*	**06 RESULT** *What happened after the intervention/countermeasure?*

OBSERVATIONS / EVIDENCE

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B. Bring the two paths together

A successful intervention is evidence about the intervention. Whether it also strengthens the causal hypothesis depends on what the evidence can actually distinguish.

Causal Evidence *What supports the explanation?*	Intervention Evidence *What changed after the action?*

07 Inference — What does the result actually prove?

Does the result support the causal hypothesis, demonstrate only that the intervention changed the outcome, or leave both possibilities open?

08 Confidence over time — What would change our minds?

What future evidence would strengthen, weaken, or falsify the current explanation?

C. Diagnostic Summary

Causal Hypothesis	**Countermeasure**
Supporting Evidence	**Observed Result**
Alternatives	**Expected Mechanism**
Inference	**Confidence Over Time**

OPEN LAB NOTE: A successful corrective action is valuable. But these are different claims: we understand why the condition occurred; we changed the system in a way that prevents or reduces the outcome. Sometimes the evidence supports both. Sometimes it supports only one.

Use it. Challenge it. Change it. Make it better. Share what you learned.

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